

The sharp Pareto entropy bound throughout the missing range in dimension one

A partial answer to arXiv:1006.2883

Autonomous proof search, lane 18

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Status. Candidate substantial partial result, likely valid. The proof settles the source question for $n = 1$ and every $\beta > 1$. The case $n \geq 2$ remains open here. Novelty is uncertain because a possibly overlapping 2024 inverse-Hölder preprint could not be inspected.

1 Source question

Bobkov and Madiman [1] prove that if a probability density on $\mathbb{R}^n$ has the form

$$f = \varphi^{-\beta},$$

where φ is positive and convex and $\beta \geq n + 1$, then fixing $\|f\|_\infty$ makes Shannon entropy maximal for the n -dimensional Pareto density. Equivalently,

$$h(f) + \log \|f\|_\infty \leq \beta \sum_{j=1}^n \frac{1}{\beta - j}. \quad (1)$$

In their final discussion they ask whether the theorem can be improved to the full normalizable range $\beta > n$. The statement is on source PDF page 13:

Let for a standard Gaussian on $\mathbb{R}$ with density g , let the random vector $Y = \sqrt{V}Z$, where V is a scalar distributed according to m , have the density g_{mix} . Then

$$\begin{aligned} \frac{n}{2} \int_0^\infty m(v) \log v \, dv &\leq h(g_{\text{mix}}) - h(g) \\ &\leq \frac{n}{2} - \log \int_0^\infty \frac{m(v)}{v} dv. \end{aligned}$$

Proof: One can write

$$g_{\text{mix}}(x) = \int_0^\infty \frac{m(v)}{(2\pi v)^{n/2}} \exp\left\{-\frac{\|x\|^2}{2v}\right\} dv,$$

since we are parametrizing using variance (rather than standard deviation). Also, $\|x\|^2/v$ is convex as a function of $(x, v) \in \mathbb{R}^n \times (0, \infty)$; indeed, the quadratic form induced by its Hessian matrix when evaluated at $(a, b) \in \mathbb{R}^n \times (0, \infty)$ is $\|ya - bx\|^2 \geq 0$. Combining this with the assumed log-concavity of $\frac{m(v)}{v^{n/2}}$, one finds that the integrand above is log-concave, and hence so is g_{mix} (by the Prékopa-Leindler inequality). Then the first inequality of Corollary IX.8 follows from concavity of entropy, while the second follows from Proposition 1.2. ■

A limitation (perhaps unavoidable) of this result is that as

the paper appears to be bearing fruit; for instance, in [13], we use some of the results in this paper as one ingredient (among several) to prove a “reverse entropy power inequality for convex measures” analogous to Milman’s reverse Brunn-Minkowski inequality [39], [41], [40], [45] for convex bodies.

We conclude with some open questions. First, the question of characterizing the κ -concavity properties of infinitely divisible laws using only knowledge of the characteristic function is an interesting one, as discussed in Section IX-C. One also hopes that Theorem 1.3 can be improved to only require $\beta > n$; this would immediately imply that many of the results in this paper stated for κ -concave measures with $\kappa \geq -1$ (or their densities) would have extended validity to general convex measures. And finally, it would be nice to use the entropic formulation of the hyperplane conjecture to improve the state-of-the-art partial results that exist.

APPENDIX A THE MULTIVARIATE PARETO DISTRIBUTION

There does not seem to be a canonical definition of a multivariate version for the Pareto distribution, although vari-

The result below proves the complete extension when $n = 1$. In particular, it settles the entire range $1 < \beta < 2$ missing from the source theorem.

2 Proof intuition

The source proof is already valid below $n + 1$ except at one inverse-Hölder step: it uses log-concavity of a normalized negative moment only on $p \geq n + 1$. In one dimension the missing extension has an elementary geometric proof. The length $L(t)$ of the sublevel interval $\{\varphi < t\}$ is a concave function of t . Every nonnegative increasing concave function on a half-line is an affine term plus a positive superposition of hinges $\min(t - 1, s - 1)$. Layer cake turns these hinges into $1 - s^{1-p}$. Consequently

$$(p - 1) \int \varphi^{-p}$$

is itself a positive concave function of $p - 1$, hence is log-concave for every $p > 1$. This is exactly the strengthened input needed by the source entropy argument.

3 A full-range one-dimensional negative-moment lemma

Lemma 1. *Let $I \subset \mathbb{R}$ be an open interval and let $\varphi : I \rightarrow (0, \infty)$ be convex. Assume that $Z(p) = \int_I \varphi(x)^{-p} dx$ is finite in the range under discussion. Then*

$$p \longmapsto (p - 1)Z(p) \tag{2}$$

is log-concave on $p > 1$.

Proof. Multiplying φ by a positive constant only multiplies (2) by a log-affine function of p , so normalize $\inf_I \varphi = 1$. Put

$$L(t) = |\{x \in I : \varphi(x) < t\}|, \quad t \geq 1.$$

The sublevel sets are intervals. If $s, t \geq 1$ and $0 < \lambda < 1$, convexity gives

$$(1 - \lambda)\{\varphi < s\} + \lambda\{\varphi < t\} \subseteq \{\varphi < (1 - \lambda)s + \lambda t\}.$$

Length is affine under Minkowski addition of intervals, so L is nonnegative, nondecreasing, and concave.

Let $c = L(1+)$ and let $a \geq 0$ be the limiting slope of L . Integrating the decreasing right derivative of L gives a positive measure ν on $(1, \infty)$ such that

$$L(t) = c + a(t - 1) + \int_{(1, \infty)} \min\{t - 1, s - 1\} \nu(ds). \tag{3}$$

The layer-cake formula and Tonelli's theorem yield

$$Z(p) = p \int_1^\infty L(t) t^{-p-1} dt.$$

The three terms in (3) can be integrated explicitly. For $p > 1$,

$$(p - 1)Z(p) = c(p - 1) + a + \int_{(1, \infty)} (1 - s^{1-p}) \nu(ds).$$

Writing $x = p - 1 > 0$, the right side is

$$cx + a + \int_{(1, \infty)} (1 - e^{-x \log s}) \nu(ds).$$

It is positive and concave in x : every integrand has nonpositive second derivative. Every positive concave function is log-concave, proving the lemma. $\square$

4 Sharp entropy bound for every $\beta > 1$

Theorem 1. *Let $\beta > 1$, let $I \subset \mathbb{R}$ be an open interval, and let $f = \varphi^{-\beta}$ be a probability density on I , where $\varphi : I \rightarrow (0, \infty)$ is convex. Then*

$$h(f) + \log \|f\|_{\infty} \leq \frac{\beta}{\beta - 1}. \quad (4)$$

The constant is sharp. Equality holds for the one-sided Pareto family, up to affine changes of variable.

Proof. The quantity on the left of (4) is invariant under invertible affine changes of the variable. Normalize $\|f\|_{\infty} = 1$. Then $\inf_I \varphi = 1$ and, with

$$Z(p) = \int_I \varphi^{-p},$$

we have $Z(\beta) = 1$.

We first record the elementary estimate used in the source proof:

$$Z(\beta + 1) \geq 1 - \frac{1}{\beta}. \quad (5)$$

Indeed, fix $y \in I$. Convexity of φ and the change of variable $z = tx + (1 - t)y$ give, for $0 < t \leq 1$,

$$t^{-1} \geq \int_I [t\varphi(x) + (1 - t)\varphi(y)]^{-\beta} dx,$$

with equality at $t = 1$. Comparing left derivatives at $t = 1$ yields

$$\beta - 1 \leq \beta\varphi(y)Z(\beta + 1).$$

Choose y with $\varphi(y)$ arbitrarily close to its infimum 1 to obtain (5).

By Lemma 1,

$$R(p) = \log((p - 1)Z(p))$$

is concave on $p > 1$. Therefore

$$\begin{aligned} R'(\beta) &\geq R(\beta + 1) - R(\beta) \\ &= \log \frac{\beta Z(\beta + 1)}{(\beta - 1)Z(\beta)} \geq 0, \end{aligned}$$

where the last inequality uses (5) and $Z(\beta) = 1$. Thus

$$\frac{1}{\beta - 1} + Z'(\beta) \geq 0.$$

Since

$$h(f) = - \int_I f \log f = \beta \int_I \varphi^{-\beta} \log \varphi = -\beta Z'(\beta),$$

we conclude that $h(f) \leq \beta/(\beta - 1)$ under the normalization $\|f\|_{\infty} = 1$. Undoing the affine change proves (4).

Finally,

$$f_{\beta}(x) = \left(1 + \frac{x}{\beta - 1}\right)^{-\beta}, \quad x > 0,$$

has mass 1, supremum 1, and entropy $\beta/(\beta - 1)$. Hence the bound is sharp. $\square$

5 Verification, higher-dimensional obstruction, and novelty

All interchanges in Lemma 1 involve nonnegative integrands, so Tonelli applies. For $\beta > 1$, convexity plus finite mass ensures the logarithmic moment is finite; alternatively the conclusion follows first by truncation and then by monotone convergence. The equality density is checked by the substitution $u = 1 + x/(\beta - 1)$.

For $n \geq 2$, the same sublevel argument only shows that $L(t)^{1/n}$ is concave. Raising the hinge representation to the n th power does not preserve the direct concavity argument. The full source question would follow from log-concavity of

$$(p-1) \cdots (p-n) \int \varphi^{-p} \quad \text{for every } p > n,$$

but this packet does not prove that assertion. The included script tests exact piecewise-affine sublevel-radius models in dimensions 2–4 and finds no counterexample; this is non-proof evidence only.

A bounded literature search used the exact title and arXiv id, the phrase “improved to only require $\beta > n$ ”, arXiv’s API, Crossref, OpenAlex, and author publication pages. No explicit later answer was found. However, a 2024 paper [2] cites an unpublished/preprint work by Bobkov, Fradelizi, Langharst, Li, and Madiman titled *When can one invert Hölder’s inequality? (and why one may want to)*. Its text was not located, and it may overlap with Lemma 1 or its higher-dimensional analogue. Novelty confidence is therefore moderate rather than high.

Human-review recommendation. Check the hinge representation (3) and compare the result with the cited inverse-Hölder preprint before asserting priority. The mathematical proof itself is short and should be reviewed as a sharp one-dimensional partial answer.

References

- [1] S. G. Bobkov and M. Madiman, *The entropy per coordinate of a random vector is highly constrained under convexity conditions*, IEEE Trans. Inform. Theory **57** (2011), 4940–4954; arXiv:1006.2883; DOI: 10.1109/TIT.2011.2158475.
- [2] M. Fradelizi, M. Madiman, A. Marsiglietti, and A. Zvavitch, *Forward and reverse entropy power inequalities in convex geometry*, Int. Math. Res. Not. **2024** (2024), 11426–11454, especially the reference to the 2024 inverse-Hölder preprint.